PathoFMPred three-representation research output

COAD TCGA-AA-A01F | TITAN, Giga-SSL and Prov-GigaPath | 2026-10-21

UNVALIDATED INTERNAL TCGA RESEARCH OUTPUT. BINARY REFERENCE RANK IS NOT PROBABILITY. DO NOT USE FOR DIAGNOSIS, TREATMENT SELECTION OR CLINICAL RISK ESTIMATION.

Output inventory

Representation	Continuous outputs	Binary outputs	Slides pooled	Output rule
TITAN	15	16	1	Q2 >= 0.20; AUROC >= 0.60
Giga-SSL	17	13	1	Q2 >= 0.20; AUROC >= 0.60
Prov-GigaPath	18	13	1	Q2 >= 0.20; AUROC >= 0.60

Binary-call overview across the three inputs

Model	Genome doubling	APC	KRAS	PIK3CA	TP53 mutation	Any called fusion	MSI-H	Strict MSI-H	Cell Cycle	HIPPO	MYC	NOTCH	PI3K	RTK RAS	TGF-Beta	TP53 pathway
TITAN	Absent	Wild type	Wild type	Wild type	Wild type	Not altered	MSI-H	MSI-H	Altered	Not altered	Not altered	Altered	Altered	Altered	Not altered	Not altered
Giga-SSL	Present	Wild type	Wild type	Wild type	Wild type	No output	Not MSI-H	Not MSI-H	No output	Not altered	No output	Altered	Altered	Altered	Altered	Not altered
Prov-GigaPath	Present	Wild type	Wild type	Wild type	Wild type	No output	Not MSI-H	Not MSI-H	No output	Not altered	No output	Altered	Altered	Altered	Not altered	Altered

Colour identifies an uncalibrated research class call, not the observed genotype or a probability. Exact LDA scores and matched-CV performance appear in the model tables.

Case context, not used as model input

The sigmoid resection contained an ulcerated grade-2 adenocarcinoma that had crossed the muscular wall into pericolic fat (pT3). Both resection ends were free of carcinoma; lymphatic invasion was recorded, and 2 of 30 regional nodes contained metastasis (pN1).

Pathology was paraphrased from TCGA-Slide-Reports.csv distributed with the TITAN study (Ding et al., Nature Medicine 2025; doi:10.1038/s41591-025-03982-3).

How to read the report

- PathoFMPred lists representation-specific fitted objects that crossed the recorded internal performance threshold and passed the package's default sample-size rule. The inventory is not an exhaustive molecular profile.

- Continuous predictions remain in the source endpoint units. Linear predictions are unconstrained and may fall outside the source assay range. The internal TCGA rank provides display context and is not a clinical reference interval.

- Binary calls use a training-derived threshold on an uncalibrated LDA score. Neither the score nor its TCGA rank is a probability.

- The Q2, AUROC and PR-AUC values below are from the primary matched-cohort nested cross-validation; they are not five-repeat averages.

- This TCGA patient had source labels for these returned objects: TITAN 10 continuous/1 binary; Giga-SSL 16 continuous/1 binary; Prov-GigaPath 16 continuous/1 binary. Where a source label was available, the patient contributed to final full-data fitting. These predictions are not held-out case validation.

TITAN: outputs meeting internal criteria

The following tables list 15 continuous and 16 binary research outputs returned for this patient and representation.

Continuous internally derived estimates

Family	Endpoint	Prediction	Units	TCGA rank	Q2 (CV)	Spearman	n	Grouped sensitivity
Aneuploidy	Aneuploidy score (Taylor)	8.211729	source endpoint units	27.6	0.296	0.572	407	effect threshold retained under grouping
MSI	MANTIS score	0.430210	source endpoint units	55.0	0.205	0.440	390	effect threshold not retained under grouping
MSI	MSIsensor score	8.991055	source endpoint units	74.8	0.219	0.312	410	effect threshold not retained under grouping
Rna Pathway Activity	Allograft rejection	-0.072250	source endpoint units	41.7	0.241	0.494	423	effect threshold not retained under grouping
Rna Pathway Activity	HALLMARK_INFLAMMATORY_RESPONSE	-0.048609	source endpoint units	43.6	0.202	0.473	423	effect threshold not retained under grouping
Rna Pathway Activity	Interferon-gamma response	0.059420	source endpoint units	60.9	0.243	0.505	423	effect threshold not retained under grouping
Rna Pathway Activity	Myogenesis	-0.357051	source endpoint units	6.5	0.216	0.449	423	effect threshold not retained under grouping
Immune/context	Aneuploidy Score (Thorsson)	7.028503	source endpoint units	24.9	0.274	0.548	407	effect threshold retained under grouping
Immune/context	Indel Neoantigens	4.122267	log1p-transformed source units	73.4	0.220	0.387	322	effect threshold not retained under grouping
Immune/context	Leukocyte Fraction	0.182788	source endpoint units	51.3	0.229	0.487	418	effect threshold not retained under grouping
Immune/context	Lymphocyte Infiltration Signature Score	-0.415420	source endpoint units	26.1	0.294	0.543	416	effect threshold not retained under grouping
Immune/context	Nonsilent Mutation Rate	1.814873	log1p-transformed source units	72.3	0.257	0.312	345	effect threshold not retained under grouping
Immune/context	SNV Neoantigens	4.704855	log1p-transformed source units	75.9	0.270	0.382	350	effect threshold not retained under grouping
Immune/context	Silent Mutation Rate	1.096294	log1p-transformed source units	70.0	0.225	0.350	345	effect threshold not retained under grouping
Immune/context	TIL Regional Fraction	9.856207	source endpoint units	55.6	0.466	0.711	413	effect threshold not retained under grouping

Binary research-model calls

Family	Endpoint	Predicted class	LDA score	TCGA rank	AUROC (CV)	PR-AUC (CV)	Training n (+/-)	Grouped sensitivity
Aneuploidy	Genome doubling	Absent	-2.424	30.1	0.752	0.668	407 (165/242)	effect threshold retained under grouping
Mutation	APC	Wild type	-2.982	17.9	0.838	0.792	382 (196/186)	effect threshold retained under grouping
Mutation	KRAS	Wild type	-3.151	16.9	0.776	0.588	382 (126/256)	effect threshold retained under grouping
Mutation	PIK3CA	Wild type	-2.243	30.2	0.637	0.287	382 (83/299)	effect threshold not retained under grouping
Mutation	TP53	Wild type	-3.171	15.3	0.859	0.811	382 (161/221)	effect threshold retained under grouping
Fusion	Any called fusion	Not altered	-0.308	22.8	0.619	0.638	257 (144/113)	effect threshold not retained under grouping
MSI	MSI-H (MANTIS >0.4)	MSI-H	-1.374	75.5	0.823	0.601	390 (73/317)	effect threshold retained under grouping

Family	Endpoint	Predicted class	LDA score	TCGA rank	AUROC (CV)	PR-AUC (CV)	Training n (+/-)	Grouped sensitivity
MSI sensitivity	MSI-H strict (MANTIS >0.6)	MSI-H	-4.323	69.0	0.900	0.737	373 (56/317)	effect threshold retained under grouping
Pathway	Cell Cycle	Altered	1.567	84.4	0.615	0.696	253 (147/106)	effect threshold not retained under grouping
Pathway	HIPPO	Not altered	-1.635	69.5	0.675	0.372	323 (63/260)	effect threshold retained under grouping
Pathway	MYC	Not altered	-1.270	58.4	0.614	0.344	323 (79/244)	effect threshold not retained under grouping
Pathway	NOTCH	Altered	1.008	87.2	0.628	0.445	323 (107/216)	effect threshold not retained under grouping
Pathway	PI3K	Altered	-0.149	54.6	0.604	0.529	323 (148/175)	effect threshold retained under grouping
Pathway	RTK RAS	Altered	2.069	65.5	0.678	0.858	323 (242/81)	effect threshold retained under grouping
Pathway	TGF-Beta	Not altered	-2.416	10.7	0.620	0.398	323 (100/223)	effect threshold retained under grouping
Pathway	TP53	Not altered	0.836	46.0	0.766	0.843	323 (208/115)	effect threshold retained under grouping

Giga-SSL: outputs meeting internal criteria

The following tables list 17 continuous and 13 binary research outputs returned for this patient and representation.

Continuous internally derived estimates

Family	Endpoint	Prediction	Units	TCGA rank	Q2 (CV)	Spearman	n	Grouped sensitivity
Aneuploidy	Aneuploidy score (Taylor)	13.942400	source endpoint units	67.9	0.264	0.517	407	effect threshold not retained under grouping
MSI	MANTIS score	0.301648	source endpoint units	17.3	0.240	0.424	390	effect threshold not retained under grouping
MSI	MSIsensor score	1.407643	source endpoint units	30.6	0.251	0.325	410	effect threshold retained under grouping
Rna Pathway Activity	Allograft rejection	-0.105085	source endpoint units	37.5	0.337	0.596	423	effect threshold not retained under grouping
Rna Pathway Activity	HALLMARK_ANGIOGENESIS	-0.159869	source endpoint units	26.8	0.202	0.458	423	effect threshold not retained under grouping
Rna Pathway Activity	HALLMARK_APICAL_JUNCTION	-0.131721	source endpoint units	24.9	0.209	0.456	423	effect threshold not retained under grouping
Rna Pathway Activity	HALLMARK_COMPLEMENT	-0.061434	source endpoint units	41.7	0.270	0.554	423	effect threshold retained under grouping
Rna Pathway Activity	IL6/JAK/STAT3	0.050587	source endpoint units	62.3	0.271	0.539	423	effect threshold not retained under grouping
Rna Pathway Activity	HALLMARK_INFLAMMATORY_RESPONSE	-0.040216	source endpoint units	49.1	0.234	0.532	423	effect threshold not retained under grouping
Rna Pathway Activity	Interferon-alpha response	-0.147410	source endpoint units	38.2	0.265	0.480	423	effect threshold not retained under grouping
Rna Pathway Activity	Interferon-gamma response	-0.146656	source endpoint units	36.3	0.304	0.563	423	effect threshold not retained under grouping
Rna Pathway Activity	Myogenesis	-0.294848	source endpoint units	5.1	0.221	0.449	423	effect threshold not retained under grouping
Immune/context	Aneuploidy Score (Thorsson)	12.112042	source endpoint units	67.9	0.227	0.493	407	effect threshold not retained under grouping
Immune/context	Leukocyte Fraction	0.189199	source endpoint units	57.8	0.301	0.565	418	effect threshold retained under grouping
Immune/context	Lymphocyte Infiltration Signature Score	-0.240839	source endpoint units	38.8	0.292	0.534	416	effect threshold not retained under grouping
Immune/context	Macrophage Regulation	-0.505379	source endpoint units	39.1	0.308	0.556	416	effect threshold not retained under grouping
Immune/context	TIL Regional Fraction	10.674070	source endpoint units	65.5	0.399	0.632	413	effect threshold not retained under grouping

Binary research-model calls

Family	Endpoint	Predicted class	LDA score	TCGA rank	AUROC (CV)	PR-AUC (CV)	Training n (+/-)	Grouped sensitivity
Aneuploidy	Genome doubling	Present	0.052	65.0	0.770	0.686	407 (165/242)	effect threshold retained under grouping
Mutation	APC	Wild type	-3.817	12.7	0.820	0.780	382 (196/186)	effect threshold not retained under grouping
Mutation	KRAS	Wild type	-2.581	14.3	0.729	0.509	382 (126/256)	effect threshold not retained under grouping
Mutation	PIK3CA	Wild type	-3.293	19.0	0.666	0.337	382 (83/299)	effect threshold not retained under grouping

Family	Endpoint	Predicted class	LDA score	TCGA rank	AUROC (CV)	PR-AUC (CV)	Training n (+/-)	Grouped sensitivity
Mutation	TP53	Wild type	-4.250	10.9	0.780	0.673	382 (161/221)	effect threshold not retained under grouping
MSI	MSI-H (MANTIS >0.4)	Not MSI-H	-6.438	20.9	0.788	0.594	390 (73/317)	effect threshold retained under grouping
MSI sensitivity	MSI-H strict (MANTIS >0.6)	Not MSI-H	-8.887	34.2	0.846	0.612	373 (56/317)	effect threshold retained under grouping
Pathway	HIPPO	Not altered	-5.258	27.4	0.616	0.337	323 (63/260)	effect threshold not retained under grouping
Pathway	NOTCH	Altered	0.593	90.6	0.625	0.474	323 (107/216)	effect threshold retained under grouping
Pathway	PI3K	Altered	3.435	97.4	0.637	0.571	323 (148/175)	effect threshold retained under grouping
Pathway	RTK RAS	Altered	4.686	67.6	0.643	0.859	323 (242/81)	effect threshold retained under grouping
Pathway	TGF-Beta	Altered	-1.566	41.3	0.607	0.376	323 (100/223)	effect threshold not retained under grouping
Pathway	TP53	Not altered	-1.099	13.5	0.737	0.843	323 (208/115)	effect threshold retained under grouping

Prov-GigaPath: outputs meeting internal criteria

The following tables list 18 continuous and 13 binary research outputs returned for this patient and representation.

Continuous internally derived estimates

Family	Endpoint	Prediction	Units	TCGA rank	Q2 (CV)	Spearman	n	Grouped sensitivity
Aneuploidy	Aneuploidy score (Taylor)	13.905074	source endpoint units	65.7	0.306	0.576	407	effect threshold retained under grouping
MSI	MANTIS score	0.266389	source endpoint units	10.4	0.292	0.452	390	effect threshold not retained under grouping
MSI	MSIsensor score	2.298627	source endpoint units	38.7	0.300	0.352	410	effect threshold not retained under grouping
Rna Pathway Activity	Allograft rejection	-0.821913	source endpoint units	0.4	0.339	0.604	423	effect threshold not retained under grouping
Rna Pathway Activity	HALLMARK_COMPLEMENT	-0.357081	source endpoint units	5.1	0.251	0.551	423	effect threshold retained under grouping
Rna Pathway Activity	HALLMARK_EPITHELIAL_MESENCHYMAL_TRANSITION	-0.680478	source endpoint units	2.5	0.201	0.471	423	effect threshold not retained under grouping
Rna Pathway Activity	IL6/JAK/STAT3	-0.387290	source endpoint units	8.9	0.224	0.501	423	effect threshold not retained under grouping
Rna Pathway Activity	HALLMARK_INFLAMMATORY_RESPONSE	-0.454514	source endpoint units	6.3	0.254	0.547	423	effect threshold not retained under grouping
Rna Pathway Activity	Interferon-alpha response	-0.499994	source endpoint units	3.9	0.275	0.515	423	effect threshold retained under grouping
Rna Pathway Activity	Interferon-gamma response	-0.621608	source endpoint units	2.2	0.332	0.586	423	effect threshold retained under grouping
Immune/context	Aneuploidy Score (Thorsson)	11.692715	source endpoint units	64.3	0.274	0.544	407	effect threshold retained under grouping
Immune/context	Fraction Altered	0.490359	source endpoint units	67.9	0.217	0.506	413	effect threshold not retained under grouping
Immune/context	Leukocyte Fraction	0.163902	source endpoint units	38.2	0.259	0.533	418	effect threshold retained under grouping
Immune/context	Lymphocyte Infiltration Signature Score	-1.612028	source endpoint units	0.1	0.275	0.549	416	effect threshold not retained under grouping
Immune/context	Macrophage Regulation	-1.534324	source endpoint units	0.1	0.334	0.582	416	effect threshold not retained under grouping
Immune/context	SNV Neoantigens	4.248818	log1p-transformed source units	51.9	0.242	0.327	350	effect threshold retained under grouping
Immune/context	TIL Regional Fraction	2.016496	source endpoint units	14.9	0.508	0.688	413	effect threshold not retained under grouping
Immune/context	Th1 Cells	-615.599993	source endpoint units	80.2	0.240	0.513	416	effect threshold not retained under grouping

Binary research-model calls

Family	Endpoint	Predicted class	LDA score	TCGA rank	AUROC (CV)	PR-AUC (CV)	Training n (+/-)	Grouped sensitivity
Aneuploidy	Genome doubling	Present	-0.515	52.7	0.779	0.712	407 (165/242)	effect threshold retained under grouping
Mutation	APC	Wild type	-8.275	4.3	0.822	0.775	382 (196/186)	effect threshold retained under grouping
Mutation	KRAS	Wild type	-3.535	19.0	0.746	0.538	382 (126/256)	effect threshold not retained under grouping
Mutation	PIK3CA	Wild type	-7.292	4.3	0.641	0.294	382 (83/299)	effect threshold not retained under grouping

Family	Endpoint	Predicted class	LDA score	TCGA rank	AUROC (CV)	PR-AUC (CV)	Training n (+/-)	Grouped sensitivity
Mutation	TP53	Wild type	-4.388	13.0	0.797	0.688	382 (161/221)	effect threshold not retained under grouping
MSI	MSI-H (MANTIS >0.4)	Not MSI-H	-3.124	66.3	0.793	0.587	390 (73/317)	effect threshold retained under grouping
MSI sensitivity	MSI-H strict (MANTIS >0.6)	Not MSI-H	-8.730	26.4	0.909	0.703	373 (56/317)	effect threshold retained under grouping
Pathway	HIPPO	Not altered	-3.113	44.1	0.632	0.308	323 (63/260)	effect threshold retained under grouping
Pathway	NOTCH	Altered	4.574	99.5	0.638	0.454	323 (107/216)	effect threshold retained under grouping
Pathway	PI3K	Altered	0.512	65.2	0.624	0.576	323 (148/175)	effect threshold retained under grouping
Pathway	RTK RAS	Altered	9.668	94.6	0.702	0.857	323 (242/81)	effect threshold retained under grouping
Pathway	TGF-Beta	Not altered	-2.440	28.6	0.632	0.424	323 (100/223)	effect threshold retained under grouping
Pathway	TP53	Altered	2.330	66.7	0.710	0.791	323 (208/115)	effect threshold retained under grouping

Reference labels and performance estimates

Q2, AUROC and PR-AUC in these tables are primary matched-cohort nested-CV estimates, not means of five repeats. The patient-level values and class calls come from final models fitted to all eligible labelled TCGA patients. The example patients contributed to fitting a final object whenever its reference label is available; these case predictions are not held-out validation. Grouped sensitivity was evaluated for all three representations by separating TCGA tissue-source-site codes in both outer and inner folds. The code is not a scanner, laboratory or independent institution identifier.

Feature group	Reference label source
Mutations	TCGA MC3 protein-altering calls within the Bailey tissue-specific driver catalogue
Aneuploidy score (Taylor)	Arm-level gain/loss burden from Taylor et al., Cancer Cell 2018, Table S2 (doi:10.1016/j.ccell.2018.03.007)
Aneuploidy Score (Thorsson)	Separate genomic-context score distributed in the PanImmune matrix of Thorsson et al., Immunity 2018 (doi:10.1016/j.immuni.2018.03.023)
Genome doubling	Binary copy-number call from Taylor et al., Cancer Cell 2018, Table S2
MSI	cBioPortal TCGA PanCancer MANTIS and MSIsensor variables
Oncogenic pathways	Sanchez-Vega pathway alteration matrix
Immune and genomic context	Published Thorsson PanImmune computational phenotypes